

	Success	Efficiency			Interaction quality					
	In-time	Action	Conversation	Overall	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness	Overall
Assistant										
Frontier API models										
Gemini 3.8 Flash	21.4	31.1	84.6	58.2	82.7	83.0	67.6	62.9	79.0	46.2
Gemini 3.1 Pro	17.6	32.7	<u>87.5</u>	60.8	71.4	78.0	61.3	67.8	68.6	30.8
GPT-6 Astra	15.8	37.9	91.3	65.1	76.9	84.0	56.6	79.0	72.8	34.5
Real-time streaming API models										
GPT-Realtime 2.1	19.0	36.5	80.2	58.9	82.1	87.6	<u>66.6</u>	<u>77.4</u>	54.1	37.1
Gemini 3.8 Live	18.1	39.2	82.1	60.9	<u>89.9</u>	76.4	62.3	52.3	80.9	42.1
Open-weight non-streaming VLMs										
Qwen3.8-27B	25.2	32.9	85.6	58.7	68.7	78.7	62.8	67.2	58.5	28.5
Kimi-VL-A3B-Thinking	<u>24.3</u>	34.1	83.7	58.1	56.4	60.0	42.4	37.1	30.8	12.0
GLM-4.6V-Flash	20.3	31.1	86.7	59.8	91.1	67.7	54.9	46.4	77.1	<u>42.6</u>
InternVideo3-8B	18.2	36.7	79.6	56.9	64.0	61.7	44.9	24.4	44.4	18.4
Muse Glimmer	15.7	39.2	75.7	58.6	70.3	<u>85.1</u>	62.6	75.9	45.2	26.4
Molmo2-8B	2.6	<u>40.9</u>	80.2	<u>65.0</u>	72.2	55.8	42.0	20.3	41.6	17.0
Open-weight streaming VLMs										
Proact-VL	20.2	37.9	63.6	50.0	69.7	30.1	22.0	20.5	25.5	16.3
StreamingVLM	19.0	29.8	81.5	54.7	76.7	34.4	29.4	29.0	19.5	23.4
MiniCPM-o 4.5	15.4	34.8	55.4	44.3	61.7	58.5	35.5	39.1	29.4	16.2
Qwen3-Omni-30B	13.4	45.9	65.5	55.1	75.7	83.1	60.6	59.5	<u>79.1</u>	30.2
Reference conditions										
No assistance	15.5	20.0	100.0	57.5	—	—	—	—	—	—

Table 2 Visual assistance in VHSim. All scores are on a 0–100 scale (↑). Success and efficiency are three-run means; efficiency is measured on in-time successes, with persona-weighted overall scores. Rubrics use run 1 and the full 16-item overall score. Bold/underline indicate the best/second-best assistant in each column; dashes denote unavailable scores.